Local Methods in Unconstrained Minimization

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1 Local Methods in Unconstrained Minimization

1.1 Geometric Foundations

Before discussing algorithms, we try to define the geometric landscape of the optimization problem $\min_{x\in\mathbb{R}^n} f(x)$:

- The **Sublevel Set** of f at value α: $\mathcal{L}_f(\alpha) = \{x \in \mathbb{R}^n \mid f(x) \le \alpha\}$.
- The **set of tangent directions** $S_f(\bar{x})$ to the level set surface at $\bar{x}$:

$$S_f(\bar{x}) = \left\{ s \in \mathbb{R}^n \mid s = \lim_{k\to\infty} \frac{y_k - \bar{x}}{\|y_k - \bar{x}\|}, \{y_k\} \to \bar{x},\ y_k \neq \bar{x},\ f(y_k) = f(\bar{x}) \right\}.$$

With these concepts, we have the following fundamental theorem:

Theorem 1.1 – Gradient is Orthogonal to Level Set

If f is differentiable at $\bar{x}$ and $s \in S_f(\bar{x})$, then $\langle \nabla f(\bar{x}), s\rangle = 0$.

Proof. For a sequence defining $s \in S_f(\bar{x})$, we have $f(y_k) = f(\bar{x})$. Using the first-order Taylor expansion:

$$0 = f(y_k) - f(\bar{x}) = \langle \nabla f(\bar{x}), y_k - \bar{x}\rangle + o(\|y_k - \bar{x}\|).$$

Dividing by $\|y_k - \bar{x}\|$ and taking the limit $k \to \infty$:

$$\lim_{k\to\infty} \left\langle \nabla f(\bar{x}), \frac{y_k - \bar{x}}{\|y_k - \bar{x}\|} \right\rangle + \frac{o(\|y_k - \bar{x}\|)}{\|y_k - \bar{x}\|} = \langle \nabla f(\bar{x}), s\rangle = 0.$$

■

1.2 Optimality Conditions

How do we recognize a local minimum x^*?

Theorem 1.2 – First-Order Necessary Condition

Let $f \in C^1$ in a neighborhood of x^*. If x^* is a local minimizer of f, then

$$\nabla f(x^*) = 0.$$

Proof. Assume by contradiction that $\nabla f(x^*) \neq 0$. Define $d := -\nabla f(x^*)$ and consider the function

$$\phi(t) := f(x^* + td),$$

for t sufficiently small. Since $f \in C^1$, ϕ is differentiable and

$$\phi'(0) = \nabla f(x^*)^\top d = -\|\nabla f(x^*)\|^2 < 0.$$

Hence, there exists $\varepsilon > 0$ such that for all $t \in (0, \varepsilon)$,

$$\phi(t) < \phi(0),$$

that is,

$$f(x^* + td) < f(x^*),$$

which contradicts the local minimality of x^*. Therefore, $\nabla f(x^*) = 0$. ■

Theorem 1.3 – Second-Order Necessary Condition

Let $f \in C^2$ in a neighborhood of x^*. If x^* is a local minimizer of f, then

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \succeq 0.$$

Proof. By Theorem 1.2, $\nabla f(x^*) = 0$. For any direction $v \in \mathbb{R}^n$, define

$$\psi(t) := f(x^* + tv).$$

Since $f \in C^2$, Taylor's theorem yields

$$\psi(t) = \psi(0) + \psi'(0)t + \frac{1}{2}\psi''(0)t^2 + o(t^2) = f(x^*) + \frac{1}{2}t^2\, v^\top \nabla^2 f(x^*) v + o(t^2),$$

where $\psi'(0) = \nabla f(x^*)^\top v = 0$. If there exists a vector v such that

$$v^\top \nabla^2 f(x^*) v < 0,$$

then for sufficiently small $|t|$ the quadratic term dominates the remainder and implies $\psi(t) < \psi(0)$, contradicting the local minimality of x^*. Hence,

$$v^\top \nabla^2 f(x^*) v \geq 0 \quad \forall v \in \mathbb{R}^n,$$

which is equivalent to $\nabla^2 f(x^*) \succeq 0$. ■

Theorem 1.4 – Second-Order Sufficient Condition

Let $f \in C^2$ in a neighborhood of x^*. If

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \succ 0,$$

then x^* is a strict local minimizer of f.

Proof. Since $\nabla^2 f(x^*) \succ 0$, there exists $m > 0$ such that

$$v^\top \nabla^2 f(x^*) v \geq m\|v\|^2 \quad \forall v \in \mathbb{R}^n.$$

By Taylor's theorem,

$$f(x^* + h) = f(x^*) + \frac{1}{2} h^\top \nabla^2 f(x^*) h + o(\|h\|^2) \geq f(x^*) + \frac{m}{2}\|h\|^2 + o(\|h\|^2).$$

For sufficiently small $\|h\|$, the remainder term is dominated by $\frac{m}{4}\|h\|^2$, implying

$$f(x^* + h) \geq f(x^*) + \frac{m}{4}\|h\|^2,$$

which proves that x^* is a strict local minimizer. ∎

Other Well-Known Optimality Conditions.

- **Convex optimality condition.** If f is convex and differentiable, then

$$\nabla f(x^*) = 0 \quad \Longleftrightarrow \quad x^* \text{ is a global minimizer.}$$

- **Higher-order necessary conditions.** If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is singular, higher-order derivatives along directions in the null space of the Hessian may be required to characterize the nature of the stationary point.

- **Nonsmooth optimality (subgradient condition).** For a convex, possibly nondifferentiable function f,

$$0 \in \partial f(x^*)$$

  is necessary and sufficient for x^* to be a global minimizer.

- **Karush–Kuhn–Tucker (KKT) conditions.** For constrained optimization problems, under appropriate constraint qualifications, any local minimizer satisfies stationarity, primal feasibility, dual feasibility, and complementary slackness conditions.

- **Second-order conditions for constrained problems.** For a twice continuously differentiable constrained problem, a KKT point with its specified multipliers is a strict local minimizer under the standard second-order sufficient condition: the Lagrangian Hessian is positive on every nonzero direction in the critical cone. The critical cone is not generally a linear subspace.

- **Strong convexity.** A finite continuous μ-strongly convex function on $\mathbb{R}^n$ is coercive and has a unique minimizer. If it is also twice differentiable, its Hessian satisfies $\nabla^2 f(x) \succeq \mu I$. For other domains, existence requires separate assumptions.

1.3 The Gradient Method

Before introducing algorithms, we first try to quantify how much the function changes.

Definition 1.1 $C_L^{1,1}$ *Function*

A function $f \in C_L^{1,1}(\mathbb{R}^n)$ if its gradient is Lipschitz continuous with constant L:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y.$$

For $C_L^{1,1}$ functions, we have a fundamental lemma:

lemma 1.1 – The Descent Lemma

If $f \in C_L^{1,1}(\mathbb{R}^n)$, then for any x, y:

$$|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2}\|y - x\|^2.$$

Proof. Put $d = y - x$. By the fundamental theorem of calculus,

$$f(y) - f(x) - \langle \nabla f(x), d \rangle = \int_0^1 \langle \nabla f(x + \tau d) - \nabla f(x), d \rangle \, d\tau.$$

Cauchy–Schwarz and gradient Lipschitz continuity give

$$|f(y) - f(x) - \langle \nabla f(x), d \rangle| \leq \int_0^1 L\tau \|d\|^2 \, d\tau = \frac{L}{2}\|y - x\|^2.$$

■

This lemma is the cornerstone of first-order methods. It guarantees that the function $f(x)$ is always bounded above by a quadratic "cup" with curvature L. Minimizing this cup gives the Gradient Descent step.

Gradient Descent

1: **Choose** $x_0 \in \mathbb{R}^n$.
2: **Iterate** $x_{k+1} = x_k - h_k \nabla f(x_k)$, with step size $h_k \geq 0$, $k = 0, 1, \ldots$

Stepsize Strategies.

- Constant step: $h_k = h > 0$.
- Diminishing step: $h_k = \frac{h}{\sqrt{k+1}}$.
- Full relaxation (completely theoretical): $h_k = \arg\min_{h \geq 0} f(x_k - h\nabla f(x_k))$.
- Armijo backtracking: decrease a trial step geometrically until

  $$f(x_k - h\nabla f(x_k)) \leq f(x_k) - \alpha h \|\nabla f(x_k)\|^2, \qquad 0 < \alpha < 1.$$

  This is a one-sided sufficient-decrease condition.
- The two-sided Goldstein rule: Find $x_{k+1} = x_k - h\nabla f(x_k)$ with $h > 0$ such that

  $$\alpha \langle \nabla f(x_k), \, x_k - x_{k+1} \rangle \leq f(x_k) - f(x_{k+1}),$$
  $$\beta \langle \nabla f(x_k), \, x_k - x_{k+1} \rangle \geq f(x_k) - f(x_{k+1}),$$

  where $0 < \alpha < \beta < 1$ are some fixed parameters. Geometric interpretation:

  $$\phi(h) := f(x - h\nabla f(x)), \quad h \geq 0$$
  $$\phi_1(h) := f(x) - \alpha h \|\nabla f(x)\|^2, \; \phi_2(h) := f(x) - \beta h \|\nabla f(x)\|^2$$
  $$\phi_2(h) \leq \phi(h) \leq \phi_1(h)$$

Theorem 1.5 – Convergence of Gradient Descent

For $f \in C_L^{1,1}$ bounded below by f^*, gradient descent with $h = 1/L$ satisfies:

$$\sum_{k=0}^{N} \|\nabla f(x_k)\|^2 \leq 2L(f(x_0) - f^*).$$

This implies $\min_{0\leq k\leq N} \|\nabla f(x_k)\| \leq O(1/\sqrt{N})$.

Proof. Apply the Descent Lemma with $y = x - h\nabla f(x)$:

$$\begin{aligned} f(y) &\leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2 \\ &= f(x) - h\|\nabla f(x)\|^2 + \frac{h^2}{2}L\|\nabla f(x)\|^2 \\ &= f(x) - h(1 - \frac{h}{2}L)\|\nabla f(x)\|^2. \end{aligned}$$

1. Full constant step. To get the best upper bound for the possible decrease of the objective function, we solve:

$$\min_{h\geq 0} \Delta(h) = \min_{h\geq 0} -h(1 - \frac{h}{2}L) \Rightarrow h^\star = \frac{1}{L}$$

Thus, one step of the Gradient Method decreases the value of the objective function at least as follows:

$$f(y) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$$

2. Constant step. Instead, if we choose $h_k = \frac{2\alpha}{L}$ with $\alpha \in (0, 1)$, then

$$f(x_k) - f(x_{k+1}) \geq \frac{2}{L}\alpha(1 - \alpha)\|\nabla f(x_k)\|^2$$

The decrease bound is maximized at $h_k = 1/L$.

3. Goldstein rule. We have

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\leq \beta\langle \nabla f(x_k), x_k - x_{k+1}\rangle = \beta h_k\|\nabla f(x_k)\|^2 \\ f(x_k) - f(x_{k+1}) &\geq h_k(1 - \frac{h_k}{2}L)\|\nabla f(x_k)\|^2 \Rightarrow h_k \geq \frac{2}{L}(1 - \beta) \\ \Rightarrow f(x_k) - f(x_{k+1}) &\geq \alpha\langle \nabla f(x_k), x_k - x_{k+1}\rangle = \alpha h_k\|\nabla f(x_k)\|^2. \end{aligned}$$

For the fixed step $1/L$, summing the one-step decrease gives

$$\frac{1}{2L}\sum_{k=0}^{N} \|\nabla f(x_k)\|^2 \leq f(x_0) - f(x_{N+1}) \leq f(x_0) - f^*.$$

This proves the stated bound. For nonzero gradients, an accepted Goldstein step also satisfies $h_k \geq 2(1 - \beta)/L$, giving a uniform decrease coefficient $2\alpha(1 - \beta)/L$. Its gradient squares are summable by the same telescoping argument. ■

1.4 Newton's Method

To get faster convergence, we need stronger assumptions.

Definition 1.2 $C_M^{2,2}$ *Function (Lipschitz Hessian)*

$f \in C_M^{2,2}$ if its Hessian is Lipschitz continuous:

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \le M\|x - y\|.$$

Newton's Method

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k).$$

Theorem 1.6 – Local quadratic convergence

Suppose $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succeq \mu I$ with $\mu > 0$, and the Hessian is M-Lipschitz on a ball $B(x^*, r)$. Take $Mr \le \mu/2$ and $x_0 \in B(x^*, r)$. Then Newton iterates remain in this ball, their Hessians are invertible, and

$$\|x_{k+1} - x^*\| \le \frac{M}{2(\mu - M\|x_k - x^*\|)} \|x_k - x^*\|^2.$$

Proof. Let $e = x_k - x^*$. Hessian Lipschitz continuity implies $\nabla^2 f(x_k) \succeq (\mu - M\|e\|)I$. Using stationarity at x^* and the integral formula for the gradient,

$$x_{k+1} - x^* = [\nabla^2 f(x_k)]^{-1} \int_0^1 [\nabla^2 f(x_k) - \nabla^2 f(x^* + te)] e \, dt.$$

The integral has norm at most $M\|e\|^2/2$, and the inverse has norm at most $1/(\mu - M\|e\|)$. This proves the bound. Since $\|e\| \le r$ and $Mr \le \mu/2$, the next error is at most $M\|e\|^2/\mu \le \|e\|/2$, so the iterates stay in the ball by induction. When $M = 0$, the next error is zero. ■

Interpretation & Insight. In this local minimization theorem, positive curvature and a sufficiently close initial point ensure quadratic convergence. The neighborhood and invertibility assumptions matter: the theorem does not provide a global guarantee for an arbitrary initial point.

References

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